Identifying Key Predictors of NCAA Tournament Upsets in March Madness

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Executive Summary

March Madness is defined by its unpredictability. Each year, lower-seeded teams defy expectations and defeat heavily favored opponents, captivating fans and destroying brackets nationwide. But are these upsets truly random, or do the data contain signals that can help identify which underdogs have a real chance?

This capstone project investigates whether advanced team performance metrics, such as offensive and defensive efficiency ratings, wins above bubble, and strength of schedule, are more predictive of NCAA Tournament upset outcomes than tournament seeding alone. Using a dataset of 578 Round of 64 matchups from the 2008 through 2026 NCAA Tournaments, this analysis compares baseline models built solely on tournament seed difference against full models that replace seeding with advanced team performance metrics from KenPom and Barttorvik, the two most widely used predictive analytics systems for ranking college basketball teams..

Key Findings:

- The best-performing model (Logistic Regression with 5 advanced metrics, independent of seed) achieved an AUC-ROC of 0.744. This represents only a marginal improvement over the seed-only baseline (0.740), suggesting that tournament seeding already captures most of the predictive information contained in advanced metrics. However, the advanced metrics model showed more meaningful gains in accuracy (+3.5 percentage points) and F1 score (+4.5%), indicating better calibration in identifying actual upsets.
- The five selected features (WAB, KenPom EM, eFG%, TOV%, OREB%) do not include tournament seed. The full model operates entirely on pre-tournament team performance data, making the comparison against the seed-only baseline a direct test of whether advanced analytics add value beyond committee-assigned seeding.

- Accuracy improved by 3.5 percentage points (64.1% vs. 60.7%) and F1 score improved by 4.5% (0.536 vs. 0.513) when advanced metrics were included.
- Wins Above Bubble (WAB) and KenPom Adjusted Efficiency Margin (KADJ EM) emerged as the most important predictors.
- The improvement, while modest, is consistent across all evaluation metrics and reflects the inherent unpredictability of single-elimination tournament play.
- An interactive prediction tool was developed that produces historically calibrated upset probabilities for any seed matchup.

1. Introduction and Research Question

1.1 Business Problem

The NCAA Men's Basketball Tournament is a single-elimination event featuring 68 teams competing across six rounds. Tournament seeding, determined by a selection committee, is the most widely used proxy for team strength. However, seeds are assigned based on subjective committee evaluation and may not fully capture the quantitative differences between teams. This creates an opportunity: if advanced statistical metrics capture team quality more precisely than seeding, they may reveal which apparent mismatches are closer than they appear.

1.2 Research Question

What combination of team-level characteristics best identifies lower-seeded teams with a higher likelihood of producing an upset in the NCAA Tournament Round of 64?

1.3 Hypothesis

Advanced team performance metrics, such as offensive and defensive efficiency, are more predictive of upset outcomes than tournament seed alone.

1.4 Scope

This analysis focuses exclusively on the Round of 64 (the first full round of the tournament), where all 8 seed matchups occur (1-vs-16 through 8-vs-9). This controlled structure ensures consistent matchup types across all years in the dataset. An upset is defined as any game in which the higher-seeded (lower-numbered) team loses to the lower-seeded (higher-numbered) team.

2. Data Overview

2.1 Dataset Description

The dataset (model_ready_r64.csv) contains 578 NCAA Tournament Round of 64 games from 2008 through 2026. Each row represents one game and includes the following categories of variables:

- **Game identifiers:** Year, winner team name, loser team name
- **Seed information:** Winner seed, loser seed, seed difference
- **Winner team metrics (W_ prefix):** 16 pre-tournament performance statistics for the winning team
- **Loser team metrics (L_ prefix):** 16 corresponding statistics for the losing team
- **Difference columns (DIFF_ prefix):** Pre-computed differences between winner and loser metrics
- **Target variable:** UPSET (binary: 1 if the higher-seeded team lost, 0 otherwise)

Total columns: 55 (48 float, 5 integer, 2 object)

2.2 Key Metrics Defined

The following advanced metrics are central to this analysis. Each is explained in plain language for readers who may not follow college basketball analytics closely.

KenPom Adjusted Efficiency Margin (KADJ EM): This measures how many points a team scores versus gives up per 100 possessions, adjusted for the strength of their opponents. Think of it as a team's net point differential on a level playing field. A rating of +20 means the team outscores opponents by 20 points per 100 possessions when controlling for who they played. A rating of +5 means they are only slightly better than average. This is widely considered the single best predictor of team quality in college basketball because it captures fundamental offensive and defensive ability independent of competition quality. Source: kenpom.com.

Barttorvik Adjusted Efficiency Margin (BADJ EM): A similar composite rating from an independent source (barttorvik.com) using a different methodology. Like KenPom, it measures adjusted offensive efficiency minus adjusted defensive efficiency. Having two independent efficiency ratings allows for cross-validation of team quality assessments.

Barthag (BARTHAG): A power rating that estimates the probability a team would beat the average Division I team on a neutral court. Values range from 0 to 1. An elite team (like a typical 1-seed) might have a Barthag of 0.97, meaning they would beat the average team 97 times out of 100. A bubble team might sit around 0.75. This metric is useful because it translates efficiency into a concrete win probability that is easy to interpret. Source: barttorvik.com.

Wins Above Bubble (WAB): This quantifies how many more (or fewer) wins a team achieved compared to what a "bubble" team (one on the edge of making or missing the tournament) would be expected to achieve against the same schedule. A WAB of +8 means the team won about 8 more games than a borderline tournament team would have with the same opponents. A WAB near 0 means the team barely earned its tournament spot. This metric is particularly useful for identifying Cinderellas because teams with low WAB are the ones who

just squeaked into the tournament, while teams with high WAB dominated their schedule. Source: barttorvik.com.

Effective Field Goal Percentage (eFG%): A shooting efficiency metric that adjusts traditional field goal percentage to account for the additional value of three-point shots. Because a made three-pointer is worth 1.5 times a made two-pointer, eFG% weights them accordingly. The formula is (FG + 0.5 x 3FG) / FGA. A team with 55% eFG% is an elite shooting team. A team with 48% eFG% is below average. This provides a more accurate picture of offensive execution than traditional field goal percentage.

Effective Field Goal Percentage Defense (eFG%D): The eFG% a team allows its opponents to shoot. Lower values indicate stronger defensive performance. A team with 45% eFG%D is elite defensively (opponents struggle to score efficiently), while 52% eFG%D indicates poor perimeter and interior defense.

Turnover Rate (TOV%): The percentage of a team's possessions that end in a turnover (a lost ball, a bad pass, a travel, etc.). A team with 15% TOV% turns the ball over on about 1 in 7 possessions. Lower is better because every turnover is a wasted scoring opportunity and often leads directly to opponent fast-break points. This metric captures ball-handling discipline and decision-making under pressure, both of which are tested in the tournament environment.

Offensive Rebound Percentage (OREB%): The percentage of available offensive rebounds a team secures after their own missed shots. A 35% OREB% means the team gets about 1 in 3 of its own misses back for another scoring attempt. Higher values indicate stronger offensive rebounding, which creates extra possessions and second-chance points. This is particularly valuable in tournament play where possessions are precious and games are often decided by small margins.

Elite Strength of Schedule (ELITE SOS): A measure of how many games a team played against elite-level opponents during the regular season. A team from a major conference (like the Big Ten or SEC) might have an Elite SOS of 30+, meaning they faced 30 or more games against high-quality opponents. A mid-major team might have an Elite SOS under 10. This metric helps explain why some teams look great statistically but have not been tested against top competition, which is relevant for predicting tournament performance.

Seed Difference (SEED_DIFF): The numerical difference between the two teams' tournament seeds. In the Round of 64, this ranges from 1 (8-vs-9 matchup, where teams are nearly equivalent) to 15 (1-vs-16 matchup, where the gap is enormous).

2.3 Data Quality

Data validation confirmed zero missing values and zero duplicate records across all 578 observations. No data cleaning was required. Outliers were identified in efficiency metrics (KADJ EM, KADJ O, KADJ D) but were retained as they represent legitimate elite teams rather than data errors.

2.4 Target Variable Distribution

Of the 578 Round of 64 games in the dataset, 153 (26.5%) resulted in upsets and 425 (73.5%) resulted in the expected outcome (favorite wins). This moderate class imbalance was addressed during modeling using SMOTE (Synthetic Minority Over-sampling Technique).

3. Methodology

3.1 Feature Engineering

The original dataset included DIFF_ columns computed as Winner minus Loser for each metric. During model development, a critical data leakage issue was identified: because the winner column contains the upset team in upset games, the sign of the difference inherently

encodes the game outcome. A model trained on these features learns to detect which team won rather than predicting whether an upset will occur.

Correction: All difference features were recomputed as Favorite minus Cinderella, where the favorite is identified by the lower seed number regardless of who won the game. This ensures that the feature values do not contain information about the outcome. The corrected features are prefixed with FIX_DIFF_ to distinguish them from the original leaky features.

Sanity check results:

- Non-upset games (favorite won): Mean FIX_DIFF_KADJ EM = +15.42 (positive, as expected, because the favorite is the better team).
- Upset games (cinderella won): Mean FIX_DIFF_KADJ EM = +5.55 (also positive, because the favorite was still better on paper but lost anyway).

Both means being positive confirms that the leakage has been eliminated. The model must now learn from the magnitude of the gap rather than its sign.

3.2 Feature Selection

Initial modeling with all 16 FIX_DIFF_ features produced worse results than the seed-only baseline due to multicollinearity (many metrics measure overlapping aspects of team quality) and overfitting (16 features for 578 observations). Feature selection reduced the model to 5 variables chosen for predictive power and independence:

1. **FIX_DIFF_WAB** - Wins Above Bubble (overall team quality relative to bubble)
2. **FIX_DIFF_KADJ EM** - KenPom Efficiency Margin (composite efficiency)
3. **FIX_DIFF_EFG%** - Effective Field Goal Percentage (shooting efficiency)
4. **FIX_DIFF_TOV%** - Turnover Rate (ball security)
5. **FIX_DIFF_OREB%** - Offensive Rebound Percentage (rebounding)

These five features capture distinct dimensions of team performance: overall quality (WAB), composite efficiency (KADJ EM), shooting (eFG%), ball handling (TOV%), and physicality (OREB%).

3.3 Class Imbalance Treatment

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training set to balance the upset and non-upset classes. This technique generates synthetic examples of the minority class (upsets) by interpolating between existing upset examples, preventing the model from defaulting to always predicting the majority class (no upset).

- Before SMOTE: 433 training samples (318 no upset, 115 upset)
- After SMOTE: 636 training samples (318 no upset, 318 upset)

SMOTE was applied only to the training set. The test set remained untouched to ensure evaluation reflects real-world class distribution.

3.4 Modeling Approach

Four models were trained and compared:

1. **Logistic Regression, Baseline:** Trained on seed difference (FIX_SEED_DIFF) only
2. **Logistic Regression, Full:** Trained on the 5 selected FIX_DIFF_ features
3. **Random Forest, Baseline:** Trained on seed difference only
4. **Random Forest, Full:** Trained on the 5 selected features

All features were standardized using StandardScaler. A stratified 75/25 train-test split preserved the upset class ratio in both sets.

3.5 Evaluation Metrics

- **AUC-ROC:** Area Under the Receiver Operating Characteristic Curve. Measures the model's ability to distinguish between upsets and non-upsets across all probability thresholds. Values range from 0.5 (random guessing) to 1.0 (perfect discrimination).
- **Accuracy:** Percentage of correct predictions.
- **F1 Score:** Harmonic mean of precision and recall, providing a balanced measure of performance on the upset class specifically.

4. Results

4.1 Model Comparison

Model	Accuracy	F1 Score	AUC-ROC
LR Full (Advanced Metrics)	**0.6414**	**0.5357**	**0.7440**
LR Baseline (Seed Only)	0.6069	0.5128	0.7402
RF Full (Advanced Metrics)	0.6345	0.4536	0.6877
RF Baseline (Seed Only)	0.6069	0.5128	0.6546

The Logistic Regression model with advanced metrics achieved the highest AUC-ROC (0.744), outperforming the seed-only baseline (0.740) across all three evaluation metrics:

- **AUC-ROC improvement:** +0.4 percentage points
- **Accuracy improvement:** +3.5 percentage points (64.1% vs. 60.7%)
- **F1 Score improvement:** +4.5% (0.536 vs. 0.513)

Logistic Regression outperformed Random Forest in both the baseline and full configurations, which is expected given the relatively small sample size (578 games) and the linear nature of the relationship between stat gaps and upset probability.

4.2 Feature Importance

Random Forest feature importance analysis identified the following ranking:

Rank	Feature	Importance
1	FIX_DIFF_KADJ EM (KenPom Efficiency Margin)	0.2655
2	FIX_DIFF_WAB (Wins Above Bubble)	0.2447
3	FIX_DIFF_EFG% (Effective Field Goal %)	0.1697
4	FIX_DIFF_TOV% (Turnover Rate)	0.1667
5	FIX_DIFF_OREB% (Offensive Rebound %)	0.1535

KenPom Efficiency Margin and Wins Above Bubble emerged as the two most important predictors, together accounting for over 50% of the model's feature importance. This suggests that composite measures of team quality are more informative than individual skill metrics (shooting, turnovers, rebounding) when predicting upsets.

4.3 Cross-Validation

Five-fold stratified cross-validation was performed to assess model stability and guard against overfitting. The Logistic Regression Full model achieved a mean AUC of 0.7518 compared to 0.7238 for the Baseline, a 3.9% improvement that confirms the test set results are not an artifact of a single favorable split. Advanced metrics provide genuine predictive value when paired with Logistic Regression. In contrast, the Random Forest Full model (0.6634) underperformed its Baseline (0.6875), suggesting that tree-based models struggle to leverage the advanced metrics effectively with this sample size. These results reinforce that the choice of both

features and algorithm matters, and that the LR Full model's advantage is robust across data splits.

4.4 Interpretation

The modest improvement of advanced metrics over seed alone reflects a fundamental reality of March Madness: tournament seeding already incorporates substantial information about team quality. The selection committee evaluates many of the same factors captured by advanced metrics. The value of advanced metrics lies in identifying specific matchups where the seed gap overstates or understates the true quality difference between teams.

The bounded nature of this improvement also reflects the inherent unpredictability of single-elimination tournament play. Unlike a seven-game series, a single game contains significant variance: hot shooting nights, foul trouble, injuries, and momentum swings can override any statistical advantage. A model that claims near-perfect accuracy in this domain should be viewed with skepticism.

5. Cinderella Analysis

5.1 Upset Rate by Seed Matchup

Historical upset rates vary dramatically by seed matchup, confirming that not all upsets are created equal:

Matchup	Total Games	Upsets	Upset Rate
1 vs 16	74	2	2.7%
2 vs 15	71	7	9.9%
3 vs 14	72	8	11.1%

4 vs 13	72	15	20.8%
5 vs 12	72	28	38.9%
6 vs 11	74	33	44.6%
7 vs 10	71	27	38.0%
8 vs 9	72	33	45.8%

Small variations in game counts across matchups (ranging from 71 to 74 total games) reflect how First Four play-in games were incorporated into the dataset across different tournament years, causing slight year-to-year fluctuations in which seeds appeared in the official Round of 64.

The 5-vs-12 matchup is the most iconic upset scenario in March Madness, and the data confirm its reputation: nearly 4 in 10 result in upsets. The 6-vs-11 matchup is even more volatile at 44.6%, likely because 11-seeds often include "First Four" play-in teams that have already won a game and built momentum. The 8-vs-9 matchup is essentially a coin flip, with the lower seed winning 45.8% of the time. At this level of seeding, committee placement and tournament luck matter nearly as much as pre-tournament strength.

The 1-vs-16 matchup remains the tournament's most predictable outcome: only 2 upsets in 74 games (2.7%). The gap in quality between a 1-seed and a 16-seed is simply too large for a single game's variance to overcome in all but the most extraordinary circumstances.

5.2 Cinderella Profile

Comparing the average statistics of upset winners (Cinderellas) against expected winners reveals key differences:

Metric	Cinderella (Upset Winner)	Expected Winner (No Upset)
KenPom Adj EM	13.637	23.313
Bart Adj EM	13.365	23.079
Barthag	0.802	0.919
Wins Above Bubble	0.332	5.700
eFG%	51.693	53.009
eFG% Defense	47.781	46.807
Turnover Rate	17.876	17.048
Win %	69.754	77.813
Elite SOS	23.150	30.365

While Cinderella's trail favorites in overall efficiency metrics, the gaps in component skills are smaller than seed difference alone would suggest. The eFG% gap is only about 1.3 percentage points, meaning Cinderellas shoot nearly as well as expected winners. The defensive eFG% gap is only 1.0 percentage points. The largest gaps appear in WAB (5.4 points) and Elite SOS (7.2 points), indicating that Cinderellas tend to come from weaker conferences with fewer opportunities to play elite opponents. They are undervalued because of their schedule, not because they lack the skills to compete.

5.3 Historical Cinderella Rankings

A Cinderella Score was computed for every game in the dataset using the trained model. This score represents the model's estimated probability that the game would result in an upset based on pre-tournament statistics. Three categories were generated:

Top Cinderella Moments: Actual upsets where the model correctly identified high upset potential. These are the "smart" upsets where the data showed the underdog was better than their seed suggested. Examples include multiple 2008 and 2009 tournament upsets where mid-major teams with strong efficiency metrics defeated power-conference opponents.

Upsets the Model Did Not See Coming: Actual upsets where the model assigned low upset probability. These represent true chaos, where the outcome defied both seeding and advanced metrics. Notably, these are almost entirely 8-vs-9 matchups, which makes sense because those teams are so statistically similar that the model cannot confidently pick a side.

Near Misses: Games where the model flagged high upset potential but the favorite survived. These are the close calls that could have gone either way, representing games where the Cinderella had the statistical profile to win but did not.

6. Interactive Prediction Tool

As a stretch goal, an interactive upset prediction tool was developed using ipywidgets in Google Colab. The tool allows users to input team statistics for any hypothetical matchup and receive a real-time upset probability estimate.

Tool Features:

- Input fields for 5 key statistics (KenPom EM, WAB, eFG%, TOV%, OREB%) for both the favorite and cinderella.
- Auto-populating default values based on historical seed averages (changing the seed slider automatically updates all stat fields to reflect the typical team at that seed).

- Tiered alert system (Favorite Holds, Watch Out, Upset Likely, High Cinderella Alert).
- Key differentials displayed showing the statistical gap between teams

Calibration: The tool produces historically realistic probabilities. For example, a default 1-vs-12 matchup returns approximately 33% upset probability, consistent with the general range of mid-seed upset rates. A default 1-vs-16 matchup returns approximately 1 to 2%, consistent with the near-impossibility of that upset historically.

Practical Application: The tool can be used during Selection Sunday by pulling current-season statistics from kenpom.com and barttorvik.com and entering them for each Round of 64 matchup. Ranking all 32 games by upset probability provides a data-driven starting point for bracket construction.

7. Model Development Journey

An important aspect of this project was the iterative process of model development and correction.

Initial Model (AUC: 0.98): The first model achieved an AUC-ROC of 0.98, which appeared exceptional but raised immediate concerns upon review. Investigation revealed data leakage: the DIFF_ columns were computed as Winner minus Loser, meaning the sign of the difference encoded the game outcome. The model was detecting which team won rather than predicting upsets from pre-game information.

Leakage-Free Model with 16 Features (AUC: 0.72 to 0.74): After recomputing features as Favorite minus Cinderella (based on seed, regardless of outcome), the AUC dropped to a realistic range. However, the seed-only baseline (0.740) outperformed the full 16-feature model (0.725), indicating that multicollinearity and overfitting were degrading performance with too many correlated features.

Final Model with 5 Selected Features (AUC: 0.744): Feature selection reduced the model to 5 uncorrelated predictors capturing distinct dimensions of team quality. The full model regained its advantage over the baseline across all metrics.

This progression, from inflated results to corrected methodology to optimized feature selection, demonstrates the importance of rigorous validation and domain expertise in applied data science. A model that appears to perform perfectly should always be questioned, and iterative refinement based on expert feedback produces more trustworthy results.

8. Stretch Goals Achieved

All four stretch goals identified in the capstone plan were completed:

Stretch Goal	Status	Implementation
Compare multiple modeling techniques	Complete	Logistic Regression vs. Random Forest, both baseline and full
Test advanced metrics vs. seed-based strategies	Complete	Full model (0.744 AUC) outperforms seed-only baseline (0.740 AUC)
Identify Cinderella candidates	Complete	Cinderella profile analysis, historical rankings, radar chart, distribution curves
Build a prediction tool	Complete	Interactive widget with auto-fill defaults and tiered alerts

9. Limitations and Future Work

9.1 Limitations

- **Sample size:** 578 games across 18 tournaments provides a moderate but not large sample, particularly for rare events like 1-vs-16 upsets (only 2 occurrences).
- **Round of 64 only:** Results may not generalize to later tournament rounds where matchup dynamics differ.
- **Pre-tournament metrics only:** The model does not account for in-game factors such as injuries, foul trouble, or momentum.
- **Modest improvement:** The AUC improvement of advanced metrics over seed alone (+0.04 points) is real but small, reflecting the information already embedded in tournament seeding.
- **No temporal validation:** The model was evaluated using a random train/test split rather than training on earlier years and testing on recent ones, which would better simulate real-world deployment.

9.2 Future Work

- **Temporal validation:** Train on earlier years and test on recent tournaments to simulate real-world deployment.
- **Additional features:** Incorporate conference tournament performance, injury reports, or betting market data.
- **Later rounds:** Extend the analysis to the Round of 32, Sweet 16, and beyond.
- **Ensemble methods:** Combine Logistic Regression and Random Forest predictions for potentially improved performance.
- **Dashboard deployment:** Convert the Colab widget into a standalone web application for broader accessibility.

10. Conclusion

This analysis confirms that advanced team performance metrics provide meaningful, if modest, predictive value for NCAA Tournament upset prediction beyond tournament seeding alone. The Logistic Regression model with five selected features (WAB, KenPom EM, eFG%, TOV%, OREB%) achieved an AUC-ROC of 0.744, outperforming the seed-only baseline of 0.740 with improvements across accuracy (+3.5 percentage points) and F1 score (+4.5%).

The most important predictors were KenPom Adjusted Efficiency Margin and Wins Above Bubble, suggesting that composite measures of team quality capture information about upset potential that individual skill metrics and seeding do not fully reflect.

While no model can reliably predict March Madness (and that unpredictability is precisely what makes the tournament compelling), this analysis demonstrates that data-driven approaches can identify matchups where the conventional wisdom of seeding understates the true competitiveness of the game. The interactive prediction tool developed as part of this project provides a practical application of these findings for future tournament analysis.

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Appendix

A. Tools and Technologies

- **Programming Language:** Python 3
- **Environment:** Google Colab
- **Key Libraries:** pandas, numpy, scikit-learn, matplotlib, seaborn, imblearn, ipywidgets
- **Data Sources:** Kaggle, KenPom (kenpom.com), Barttorvik (barttorvik.com), Sports Reference

B. Code Repository

Complete code is available in the accompanying Jupyter Notebook (CapstoneModel.ipynb), which includes all data validation, exploratory analysis, modeling, Cinderella analysis, and the interactive prediction tool.